

Major Project – Internship Report

```
# -----  
# Intelligent Resume Screening System ---> By Sidharth  
# -----
```

Code 1:

```
pip install pandas scikit-learn nltk
```

Output:

```
Requirement already satisfied: pandas in c:\users\lenovo\anaconda3\lib\site-packages (2.3.3)  
Requirement already satisfied: scikit-learn in c:\users\lenovo\anaconda3\lib\site-packages (1.7.2)  
Requirement already satisfied: nltk in c:\users\lenovo\anaconda3\lib\site-packages (3.9.2)  
Requirement already satisfied: numpy>=1.26.0 in c:\users\lenovo\anaconda3\lib\site-packages  
(from pandas) (2.3.5)  
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\lenovo\anaconda3\lib\site  
-packages (from pandas) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in c:\users\lenovo\anaconda3\lib\site-packages  
(from pandas) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in c:\users\lenovo\anaconda3\lib\site-packages  
(from pandas) (2025.2)  
Requirement already satisfied: scipy>=1.8.0 in c:\users\lenovo\anaconda3\lib\site-packages  
(from scikit-learn) (1.16.3)  
Requirement already satisfied: joblib>=1.2.0 in c:\users\lenovo\anaconda3\lib\site-packages  
(from scikit-learn) (1.5.2)  
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\lenovo\anaconda3\lib\site  
-packages (from scikit-learn) (3.5.0)  
Requirement already satisfied: click in c:\users\lenovo\anaconda3\lib\site-packages (from nltk)  
(8.2.1)  
Requirement already satisfied: regex>=2021.8.3 in c:\users\lenovo\anaconda3\lib\site-packages  
(from nltk) (2025.9.1)  
Requirement already satisfied: tqdm in c:\users\lenovo\anaconda3\lib\site-packages (from nltk)  
(4.67.1)  
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-packages  
(from python-dateutil>=2.8.2->pandas) (1.17.0)  
Requirement already satisfied: colorama in c:\users\lenovo\anaconda3\lib\site-packages  
(from click->nltk) (0.4.6)
```

Note: you may need to restart the kernel to use updated packages.

Code 2:

```
import pandas as pd  
import nltk  
  
from nltk.corpus import stopwords  
from nltk.tokenize import word_tokenize  
from sklearn.feature_extraction.text import TfidfVectorizer  
from sklearn.metrics.pairwise import cosine_similarity
```

Code 3:

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

Output:

```
[nltk_data] Downloading package punkt to
[
[nltk_data]   C:\Users\Lenovo\AppData\Roaming\nltk_data...
[
[nltk_data]   Package punkt is already up-to-date!
[
[nltk_data] Downloading package punkt_tab to
[
[nltk_data]   C:\Users\Lenovo\AppData\Roaming\nltk_data...
[
[nltk_data]   Package punkt_tab is already up-to-date!
[
[nltk_data] Downloading package stopwords to
[
[nltk_data]   C:\Users\Lenovo\AppData\Roaming\nltk_data...
[
[nltk_data]   Package stopwords is already up-to-date!
```

Code 4:

```
# Loading Dataset

data = pd.read_csv("Resume.csv")

print("Dataset is Loading...\n")
print(data.head())
print(data.columns)
print("\nDataset Loaded Successfully")
```

Output:

Dataset is Loading...

	ID	Resume_str \
0	16852973	HR ADMINISTRATOR/MARKETING ASSOCIATE\...
1	22323967	HR SPECIALIST, US HR OPERATIONS ...
2	33176873	HR DIRECTOR Summary Over 2...
3	27018550	HR SPECIALIST Summary Dedic...
4	17812897	HR MANAGER Skill Highlights ...

	Resume_html	Category
0	<div class="fontsize fontface vmargins hmargin...	HR
1	<div class="fontsize fontface vmargins hmargin...	HR
2	<div class="fontsize fontface vmargins hmargin...	HR
3	<div class="fontsize fontface vmargins hmargin...	HR
4	<div class="fontsize fontface vmargins hmargin...	HR

```
Index(['ID', 'Resume_str', 'Resume_html', 'Category'], dtype='object')
```

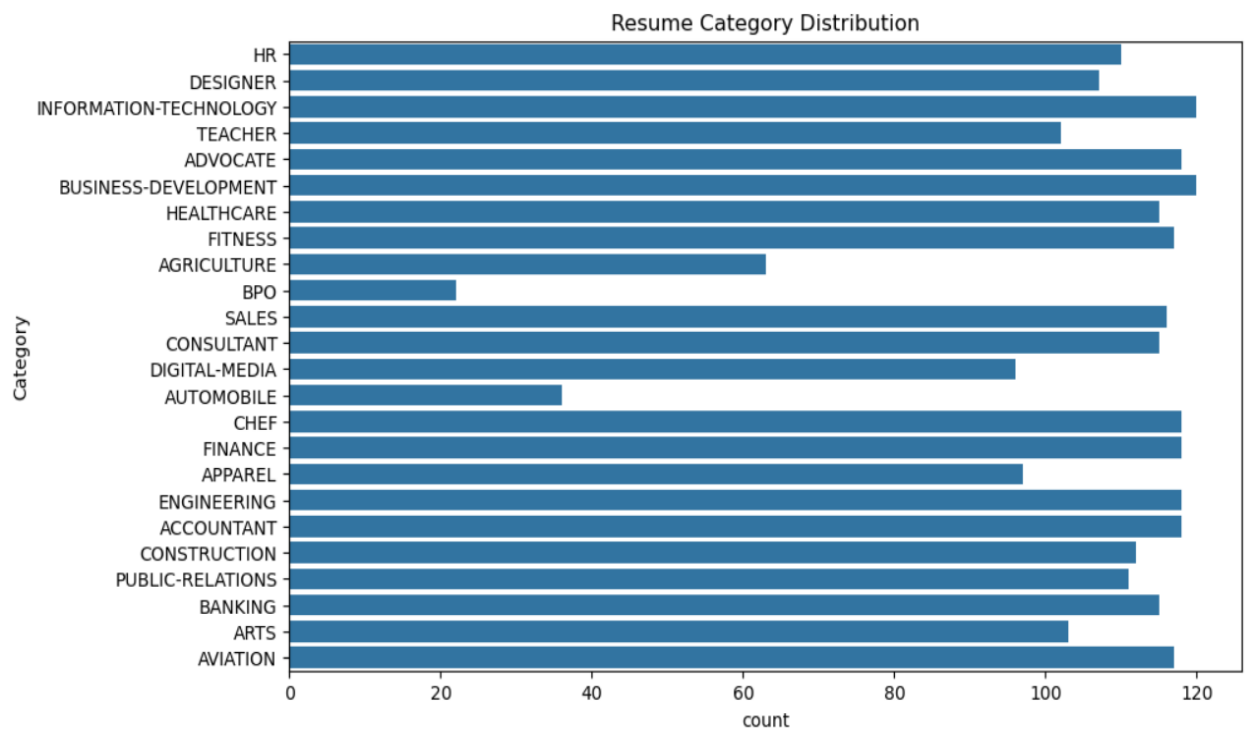
Dataset Loaded Successfully

Code 5:

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))
sns.countplot(y=data["Category"])
plt.title("Resume Category Distribution")
plt.show()
```

Output:



Code 6:

```
# Extracting Resume Text

resumes = data['Resume_str'].astype(str)
```

Code 7:

```
# Using Text Preprocessing

stop_words = set(stopwords.words('english'))

def preprocess_text(text):
    text = text.lower()
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    tokens = [word for word in tokens if word not in string.punctuation]
    return " ".join(tokens)
```

Code 8:

```
# Applying preprocessing to resumes

processed_resumes = resumes.apply(preprocess_text)
```

Code 9:

```
# Job Description:

job_description = "python , machine learning , data science , data analysis , pandas ,
                  numpy , scikit-learn , data visualization"

processed_job = preprocess_text(job_description)
```

Code 10:

```
vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(processed_resumes)

job_vector = vectorizer.transform([processed_job])
```

Code 11:

```
# Calculating Similarity:

similarity_scores = cosine_similarity(job_vector, tfidf_matrix)
```

Code 12:

```
# Ranking Resumes:

scores = similarity_scores.flatten()

ranking = pd.DataFrame({
    "Resume_ID": data["ID"],
    "Score": scores
})

ranking = ranking.sort_values(by="Score", ascending=False)

ranking.index.name = "Index"
```

Code 13:

```
# Show Top Resumes:

n = int(input("How many Top Resumes you want:/n"))

print("Top",n," Matching Resumes:")

print("For --> python , machine learning , data science , data analysis , pandas , numpy ,
scikit-learn , data visualization")

# print(ranking.head(n))
ranking.head(n)
```

Output:

How many Top Resumes you want: 20

Top 20 Matching Resumes:

For --> python , machine learning , data science , data analysis , pandas , numpy , scikit-learn , data visualization

	Resume_ID	Score
Index		
1339	18448085	0.264112
1218	21156767	0.244059
926	62994611	0.233886
1762	12011623	0.212316
1303	42156237	0.127768
331	18067556	0.124413
1040	12351749	0.122694
929	11813872	0.106945
2153	34953092	0.099848
1091	24610685	0.096246
315	10265057	0.083114
1348	22946204	0.080457
1259	27080812	0.073118
267	13405733	0.072205
662	14871762	0.071325
294	22776912	0.070015
2291	12777487	0.068769
316	30223363	0.068754
778	26585242	0.063449
1251	18442517	0.063388